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2025-yil 14-iyul 1-variant

1. $f(x) = x^2$ funksiyaga $(1;1)$ va $(0;0)$ nuqtalardan o'tadigan to'g'ri chiziqqa parallel urinma tenglamasini tuzing.

A) $y = x - 0{,}5$
B) $y = x - 0{,}25$
C) $y = x - 1$
D) $y = x - 1{,}25$

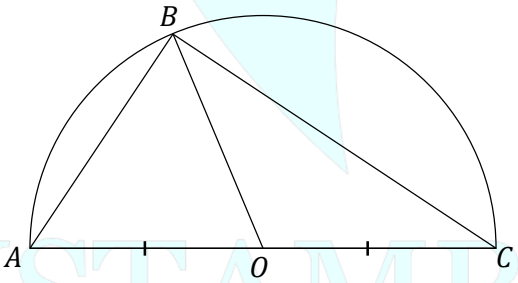
2 $\int_0^1 (x^2 - x + 2023)^{2023} \cdot (2x - 1)dx$ integralni hisoblang.

A) -1
B) 0
C) $2 \cdot 2^{2023}$
D) $2 \cdot 2^{2024}$

3. $29^{134} + 23 \cdot 24 \cdot 26 \cdot 29$ sonining oxirgi raqamini toping.

A) 6
B) 7
C) 8
D) 9

4. Radiusi $3\sqrt{2}$ ga teng bo'lgan yarim aylanaga yuzasi eng katta uchburchak ichki chizilgan. Shu uchburchak yuzasini toping.

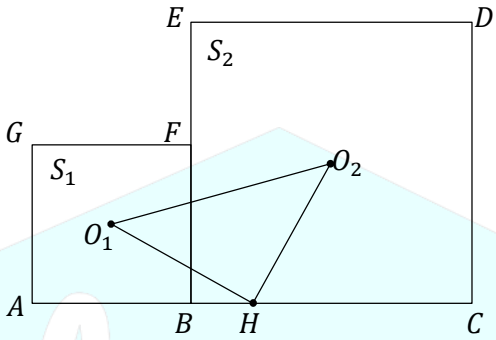


A) 18
B) 36
C) 72
D) 40

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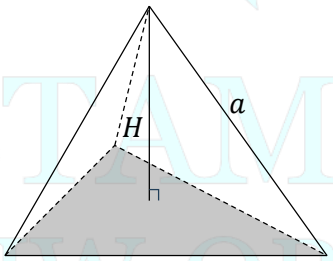
5. Rasmda $ABGF$ va $BCDE$ yuzalari S_1 va S_2 bo'lgan kvadratlar tasvirlangan va $S_1 + S_2 = 88$ ga teng. H nuqta BC tomonidan olingan bo'lib, bunda $AH = HC$bo'lsa, $S_{O_1HO_2}$ uchburchak yuzasini toping.



A) 22
B) 11
C) 20
D) 12

6. $\frac{5}{\sqrt{5}+\sqrt{3}} + \frac{5}{\sqrt{5}-\sqrt{3}}$ hisoblang.
A) $5\sqrt{5}$
B) $2\sqrt{3}$
C) $5\sqrt{3}$
D) 0

7. Rasmda tasvirlangan muntazam uchburchakli piramidaning asosining yuzi $27\sqrt{3}$ ga teng. Piramida balandligi esa yon qirrasidan ikki marta kichik. Piramida hajmini toping.



A) 27
B) 56
C) 54
D) 36

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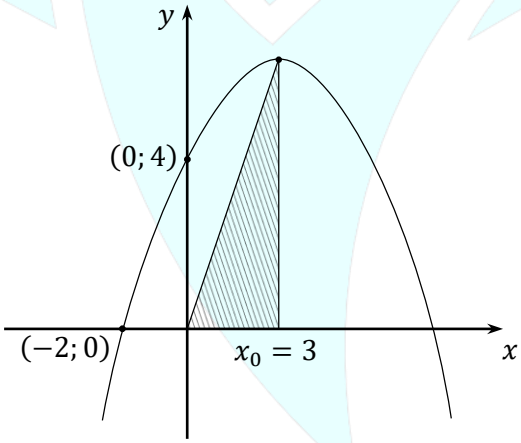
8. $\frac{5^{1+\log_6 7}\cdot 6^{\log_7 5}\cdot 7^{\log_5 6}}{6^{\log_5 7}\cdot 5^{\log_7 6}\cdot 7^{1+\log_6 5}}$ hisoblang.

A) $\frac{5}{17}$
B) $\frac{5}{7}$
C) $\frac{16}{11}$
D) $\frac{18}{19}$

9. Agar $m=6$, $n=3\frac{1}{3}$ bo'lsa, $\left(m+n-\frac{mn}{m+n}\right)\left(\frac{m}{m+n}+\frac{n}{m-n}-\frac{mn}{n^2-m^2}\right)^{-1}+n$ ni hisoblang.

A) 5
B) 7
C) 6
D) 8

10. Rasmda tasvirlangan parabolaning uchi $x_0=3$ ga teng. Bo'yalgan soha yuzini toping.



A) $\frac{75}{8}$
B) $\frac{25}{4}$
C) $\frac{19}{28}$
D) 1

11. Radiusi $2\sqrt{3}$ ga teng bo'lgan sharga eng katta hajmli silindr ichki chizilgan. Silindrning hajmini toping.

A) 24π
B) 8π
C) 16π
D) 32π

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12. $\frac{27}{13}+\frac{77}{19}-\frac{93}{23}$ ning qiymati qaysi oraliqga tegishli.

A) $(1;2)$
B) $(3;4)$
C) $(2;3)$
D) $(0;1)$

13. Agar $f(a)=(1-a)\cdot(1-2a)\cdot\ldots\cdot(1-2023a)$ bo'lsa, $\frac{f(a)\cdot\left(\frac{1}{1-a}+\frac{2}{1-2a}+\ldots+\frac{2023}{1-2023a}\right)}{f'(a)}$ ning qiymatini toping.

A) 1
B) $\frac{2a^{2023}}{35}$
C) -1
D) $-\frac{a^{2023}}{34}$

14. Agar $\alpha\in\left(\frac{\pi}{2};\pi\right)$ va $\sin\alpha=0{,}2$ bo'lsa, $\mathrm{tg}(2\pi-2\alpha)$ ni hisoblang.

A) $\frac{2\sqrt{6}}{23}$
B) $\frac{4\sqrt{6}}{23}$
C) $2\sqrt{3}$
D) $\frac{5\sqrt{6}}{41}$

15. $\int_{-1}^{1}\sqrt{1-x^2}dx$ ni hisoblang.

A) $\frac{\pi}{2}$
B) $\frac{\pi}{4}$
C) $\frac{\pi}{9}$
D) π

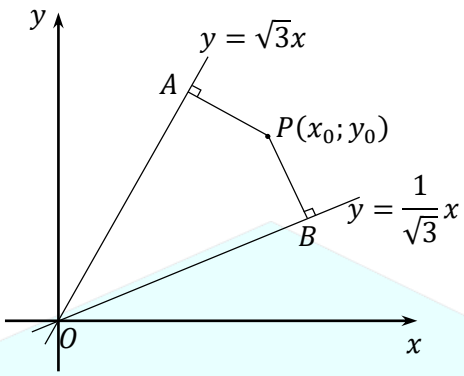
16. Soddalashtiring: $2ax-6ay+bx-3by$

A) $(2a+b)\cdot(3y-x)$
B) $(2a+b)\cdot(x-3y)$
C) $(2a-b)\cdot(3y-x)$
D) $(2a-b)\cdot(3y-x)$

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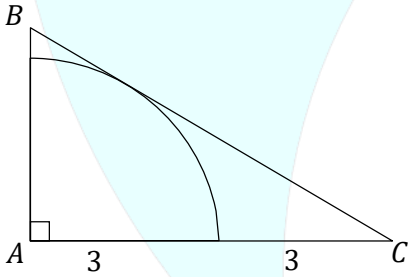
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17. Agar $x_0^2 + y_0^2 = 24$, $PB \perp OB$, $PA \perp OA$ bo'lsa, S_{OAPB} =?



A) $\frac{31\sqrt{3}}{12}$
B) 2
C) $3\sqrt{3}$
D) 6

18. Rasmda ABC to'g'ri burchakli uchburchakka chorak aylana ichki chizilgan bo'lsa, ABC uchburchakning yuzini hisoblang.

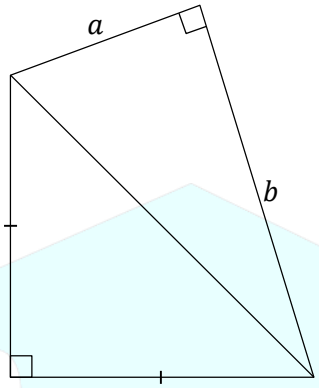


A) $6\sqrt{3}$
B) 18
C) 9
D) $12\sqrt{3}$

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19. Rasmda to'rtburchak tasvirlangan bo'lib $a + b = 10$ ga teng. To'rtburchakning yuzini toping.



A) 21
B) 25
C) 36
D) 18

20. $\cos x + \cos 2x + \cos 3x + \cos 4x = 0$ tenglama $\left[-\frac{\pi}{2}; \frac{\pi}{2}\right]$ oraliqda nechta haqiqiy ildizga ega.
A) 2 ta
B) 3 ta
C) 4 ta
D) 5 ta

21. Arifmetik progressiyada $a_1 + a_2 = 20$ va dastlabki beshta hadi yig'indisi 20 ga teng bo'lsa, dastlabki yigirmata hadi yig'indisini toping.
A) -520
B) 520
C) 500
D) -500

22. Eng katta sonni toping.
A) $\frac{5}{8}$
B) $\frac{23}{36}$
C) $\frac{47}{72}$
D) $\frac{7}{12}$

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23. $9x^6 = (4x^2 - x)^2$ tenglamaning haqiqiy ildizlari sonini toping.
A) 4 ta
B) 2 ta
C) 3 ta
D) 5 ta

24. $(x-1)(|x|+1) \leq 3$ tengsizlikni yeching.
A) $(-\infty; 2]$
B) $(-\infty; 1-\sqrt{3}]$
C) $(-\infty; \sqrt{3}]$
D) $(-\infty; -\sqrt{3}]$

25. $x^2 - 2|x| - 24 = 0$ tenglamaning ildizlar ko'paytmasini toping.
A) -24
B) 576
C) -36
D) -16

26. Do'konda 5 ta bo'sh o'rin bor. 2 ta odamni kassaga, 2 ta odamni skladga va 1 ta odamni sotuvchi qilib nechta usulda joylashtirish mumkin?
A) 30
B) 15
C) 60
D) 120

27. $y = \frac{2x-1}{x}$ ning qiymatlar sohasini toping.
A) $(-\infty; 2) \cup (2; \infty)$
B) $(-\infty; -2) \cup (2; \infty)$
C) $(-\infty; 0) \cup (2; \infty)$
D) $(-\infty; \infty)$

28. $3^{|6x-4|} = 9^{4-3x}$ tenglamaning ildizlari yig'indisini toping. (Agar bitta ildiz bo'lsa shu ildizini toping)
A) 1
B) 1,2
C) 0,2
D) 2

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29. Hisoblang: $\left(\frac{9^{\frac{3}{2}}-8^{\frac{4}{3}}}{\sqrt{11}}\right)^2$

A) 13
B) 11
C) $\sqrt{11}$
D) 1

30. Toshbaqa 1 minutda $2\,m$ yo'l bosadi. $150\,m$ ni necha soatda bosib o'tadi.
A) 1 soat
B) 1,25 soat
C) 1,5 soat
D) 2 soat

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